

2. Determine whether f is a function from $\mathbf{Z}$ to $\mathbf{R}$ if

- a) $f(n) = \pm n$.
- b) $f(n) = \sqrt{n^2 + 1}$.
- c) $f(n) = 1/(n^2 - 4)$.

a) NO

b) YES

c) YES

5. Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.

- a) the function that assigns to each bit string the number of ones in the string minus the number of zeros in the string
- b) the function that assigns to each bit string twice the number of zeros in that string
- c) the function that assigns the number of bits left over when a bit string is split into bytes (which are blocks of 8 bits)
- d) the function that assigns to each positive integer the largest perfect square not exceeding this integer

a) Domain: bit string / Range: (bit string size, 0)

b) Domain: bit strings / Range: $[2(\text{bit string size}), 0]$

c) Domain: bit string / Range: (0, 8)

d) Domain: $\mathbf{Z}^+$ / Range: $\mathbf{Z}^+$

10. Determine whether each of these functions from $\{a, b, c, d\}$ to itself is one-to-one.

- a) $f(a) = b, f(b) = a, f(c) = c, f(d) = d$
- b) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$
- c) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

11. Which functions in Exercise 10 are onto?

10a) injective, diff results for each input.

b) $f(a), f(b) = b$: not injective

c) $f(a) = d, f(b) = d$: not injective

11a) surjective, each element in the range is used

b) a is not an output,

c) a is not an output.

12. Determine whether each of these functions from $\mathbf{Z}$ to $\mathbf{Z}$ is one-to-one.

- a) $f(n) = n - 1$
- b) $f(n) = n^2 + 1$
- c) $f(n) = n^3$
- d) $f(n) = \lfloor n/2 \rfloor$

13. Which functions in Exercise 12 are onto?

a) injective

b) not injective $f(1), f(-1) = 2$

c) injective

d) not injective, $f(1), f(2) = 1$

13a) surjective, each element in the range is used

b) not surjective, no positive result

c) not surjective, no n such that $f(n) = 2$

d) surjective

20. Give an example of a function from $\mathbf{N}$ to $\mathbf{N}$ that is

- a) one-to-one but not onto.
- b) onto but not one-to-one.
- c) both onto and one-to-one (but different from the identity function).
- d) neither one-to-one nor onto.

a) $f(n) = n^2$

b) $f(n) = \lfloor n/2 \rfloor$

c) $f(n) = \begin{cases} n+1 & \text{if } n \text{ is odd} \\ n-1 & \text{if } n \text{ is even} \end{cases}$

d) $f(n) = n^3$

34. Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements.
- a) If $f \circ g$ is onto, then f must also be onto.
 - b) If $f \circ g$ is one-to-one, then g must also be one-to-one.
 - c) If $f \circ g$ is a bijection, then g is onto if and only if f is one-to-one.

a) The results of $f \circ g$ are from C .
thus, for $f \circ g$ to be surjective, then its range is C ,
 $f(B)$ must be surjective as well.

b) There can be no repeated values in the output of $f \circ g$. Thus, the inputs for f must all be distinct.
therefore, g must be injective.

c) given that $f \circ g$ is a bijection, A maps to C . This only works if both functions fully represent their sets